

H-101
Atmospheric crude charge heater
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 128 rad. tubes 6" x 18.3 m
DESIGN 38.5 barg / 455 C; CA 3 mm
DUTY 60.68 MW
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

LP STEAM HEADER
CFU-900-PR-PID-016

FROM E-111 (CIT)
CFU-100-PR-PID-003

BFW HEADER
CFU-900-PR-PID-016
1" BFW-100-072-01-H

TV-1024
FC

TIC
1024

350 C
TT
1024

PI
1227

STRIPPING STEAM HDR
CFU-100-PR-PID-006

TT
1020A-C

2003
SIF-106

TO C-101 FLASH ZONE
CFU-100-PR-PID-006

TT
1020

TIC
1020

SP CASCADE TO PIC-1021
(CFU-100-PR-PID-005); FF FROM FIC-1001

TI
1011

8" P-100-060-B3-H

TI
1012

8" P-100-060-B3-H

TI
1013

8" P-100-060-B3-H

TI
1014

8" P-100-060-B3-H

TI
1015

8" P-100-070-B3-H

TI
1016

8" P-100-070-B3-H

TI
1017

8" P-100-070-B3-H

TI
1018

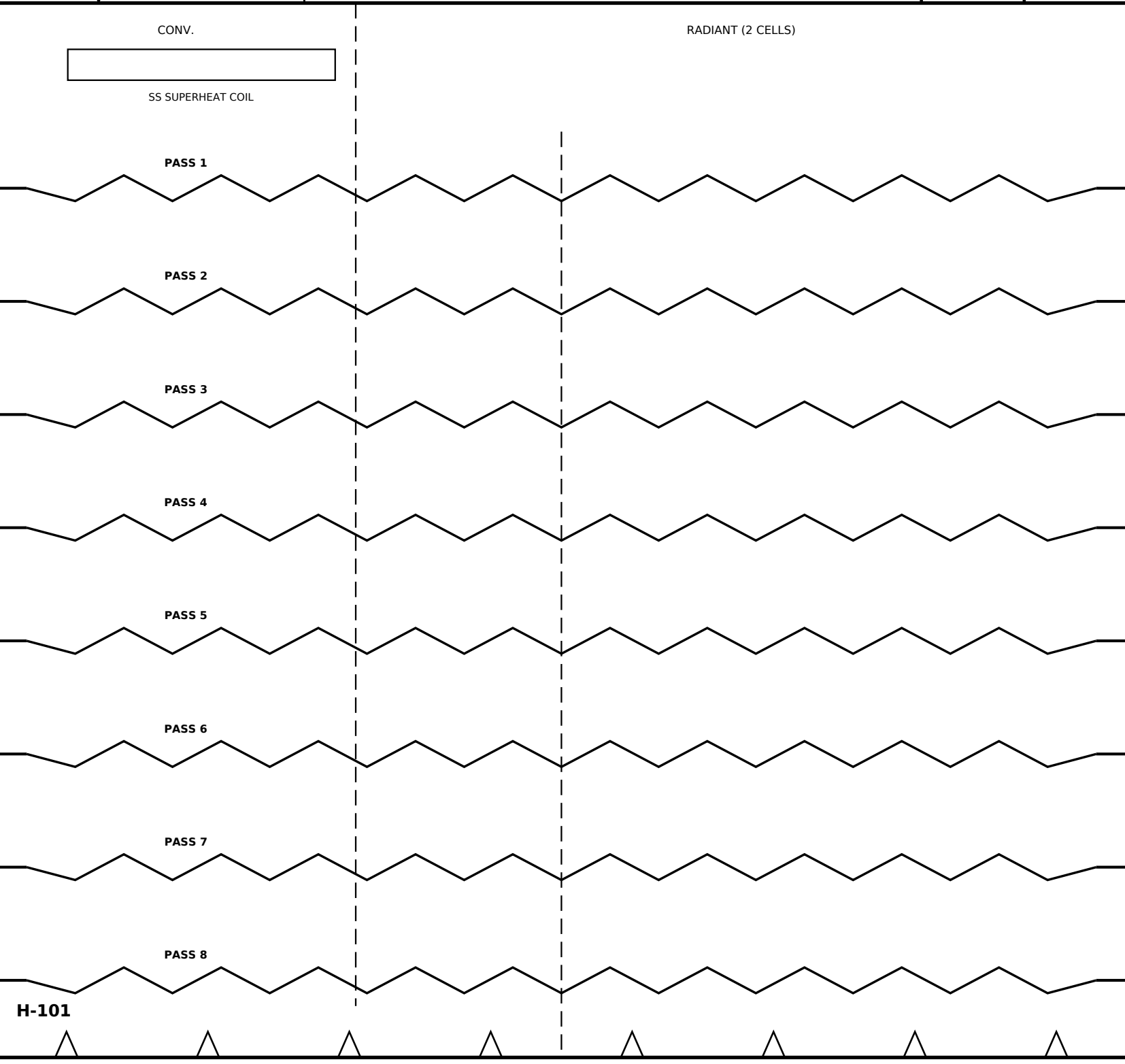
8" P-100-070-B3-H

TDIC
1019

BIAS TO FIC-1011-1018
(TOTAL FLOW HELD)

SIF-101

TO XV-1021/1022, XV-1026 (CFU-100-PR-PID-005)



- NOTES**
- Passes shown schematically (convection + radiant in series per pass); 8 identical passes.
 - Tube-skin thermocouples: 4 per pass (TI-1101..1132) - shown on heater vendor P&ID, alarmed in DCS.
 - Pass flow SIS transmitters FT-100A/B/C (2003 per pass) independent from BPCS FT-100x (SIF-101).
 - Decoking (steam-air) connections and spool pieces at inlet / outlet manifolds by heater vendor.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST		
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING & INSTRUMENT DIAGRAM		
ATMOSPHERIC HEATER H-101 - PROCESS COILS		
DISCIPLINE	STAGE	SCALE
PROCESS	FEED	NTS
		SHEET
		5 OF 17
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DRAWING NUMBER		DATE
		2026-10-02
		REV
		A